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RESEARCH ARTICLE

PlanTEA-WM: A Multi-User Web Platform for Routine Planning and Anticipating Everyday Situations in Individuals With Autism Spectrum Disorder

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ABSTRACT PlanTEA-WebMultiuser (PlanTEA-WM) is an innovative web platform designed to enhance planning, anticipating and exposing anxiogenic situations for individuals with Autism Spectrum Disorder (ASD). Conceived as an evolution of the PlanTEA mobile application, this multi-user solution facilitates the creation of visual routines through pictograms, integrating Large Language Models (LLMs) to automate and simplify the routine-generation process. PlanTEA-WM offers essential functionalities such as a shared calendar, an invitation system for collaboration among families, educators, and therapists, and plan exportation in versatile formats like XML and PDF. To improve accessibility and collaborative use, PlanTEA-WM stores user data in a remote database, supporting multi-user access from multiple devices. The platform has been preliminarily evaluated in collaboration with specialized autism associations, with results indicating high usability, accessibility, and overall acceptance. These findings highlight its potential to foster autonomy in individuals with ASD and strengthen coordination within their support environments. Within this scope, the contribution lies in combining multi-user management, LLM-assisted routine generation, and standards-based export in a single web platform designed around ASD-oriented guidelines and workflows.

INDEX TERMS Assistive technology, autism spectrum disorder (ASD), large language models (LLMs), pictograms, routine planning and anticipation, usability.

I. INTRODUCTION

Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder with a strong genetic basis that manifests from an early age. It is characterized by impairments in skill acquisition across multiple domains, stereotyped behaviors, rigid routines, repetitive and restricted interests, and

persistent deficits in socialization, reciprocal interaction, and communication [1].

The diagnosis of this disorder has increased significantly over the past 40 years [2]. According to the World Health Organization (WHO), the global prevalence is estimated at 1 case per 100 children (1%), with variations among countries depending on their diagnostic capabilities [3].

According to the criteria of the American Psychiatric Association (APA), autism spectrum disorder can manifest

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in three levels of severity, primarily defined by deficits in social and communication skills, as well as by repetitive and restrictive behavior patterns. Level 1 (“requiring support”) typically involves mild communication difficulties and challenges in adapting to changes or unstructured contexts. Level 2 (“requiring substantial support”) is characterized by more pronounced verbal communication deficits, atypical social behavior, heightened anxiety in response to change, and disruptive repetitive behaviors. Level 3 (“requiring very substantial support”) involves severely limited or unintelligible communication and highly stereotyped behaviors that significantly affect daily functioning.

In the field of intervention for individuals with ASD, the use of Augmentative and Alternative Communication (AAC) systems and visual representations, such as pictograms, is a highly valuable resource for anticipating daily activities, structuring routines, and thus fostering autonomy [4]. A pictogram is a simple visual representation that uses images to convey information or ideas in a clear and universal manner. Typically, pictograms employ standardized graphical elements that simplify the communication of messages, concepts, or instructions, facilitating immediate recognition regardless of linguistic ability or cultural background. The use of these tools has also been shown to reduce anxiety and confusion that individuals with ASD may experience when facing unfamiliar situations, changes in daily organization, or a lack of clarity in received instructions [5]. Children with ASD may face difficulties in acquiring and applying prior knowledge in anticipatory contexts, underscoring the necessity for interventions to enhance goal-directed anticipation and alleviate social and communication challenges [6].

In this context, technology has been shown to bring significant benefits to this group of users, enhancing skills such as communication, autonomy, and social interaction, among others [7], [8]. In recent years, there has been a proliferation of applications aimed at supporting individuals with ASD [9]. However, only a few provide support for routine planning and anticipation of daily tasks. Additionally, these applications present multiple limitations, as their design does not consider the specific characteristics that software applications intended for this user group should support [10]. These include a preference for visual processing, differences in sensory perception, attention to detail, a strong preference for routine and order, as well as specific sounds, objects, and people, among others [11]. As a result, these applications lack the adaptability necessary to meet the unique needs, preferences, and characteristics of each user with ASD [12]. Due to these shortcomings, families and caregivers still resort to traditional methods (paper, laminators and adhesive tape) to plan and anticipate daily situations.

To address this need, PlanTEA [13] was developed as an Android mobile application to support parents, therapists, and healthcare professionals in their *planner* role, when designing activity sequences within clinical settings. It offers a straightforward pictogram-based interface to enhance anticipation and communication. This application is intended

to assist ASD individuals across different levels of need allowing *planners* control step granularity and display modes (image-only, text-only, or both).

Therefore, PlanTEA supports two types of users or roles: the *planner* and the individual with ASD. The *planner* is responsible for creating and managing plans. A plan is an ordered sequence of pictograms, created through direct manipulation techniques, specifically *drag & drop*, an interaction technique that allows users to move elements within a graphical interface by dragging them with the cursor and dropping them in another location [14]. Additionally, the application supports flexible plan editing enabling plans to be simplified (by reducing the number or length of steps) or expanded as needed. Generic pictograms can also be replaced with real photographs to enhance familiarity. At the same time, individuals with ASD can use the plans to anticipate situations through visualization, simulation, and progress tracking.

The design and evaluation of PlanTEA [15] involved ASD experts from various Spanish associations, including FESPAU¹ (Spanish Federation of Autism), AUTRADE² (Regional Association for Autism and Other Developmental Disorders), and FACLM³ (Autism Federation of Castilla-La Mancha). Although the evaluation yielded highly positive results, the app’s single-user design and reliance on local storage constrained collaboration and limited flexible access to plans by different stakeholders. These shortcomings underscored the need for a multi-user solution capable of addressing the practical demands of parents, therapists, and educators, who require coordinated, device-independent use with remote and simultaneous access.

To overcome these limitations, PlanTEA-WebMultiuser (PlanTEA-WM) was developed as a web-based platform that retains the benefits of PlanTEA while extending its accessibility and functionality. It incorporates Artificial Intelligence (AI) and advanced tools, such as text-to-pictogram translation and plan export to standard formats (e.g., XML, PDF), to better serve individuals with ASD—regardless of severity level—and their planners. In particular, the system leverages Large Language Models (LLMs), which are a type of generative AI system trained on extensive textual data. These models enable the automatic generation of personalized visual routines from natural language input.

Recent overviews on AI and ASD-oriented software have shown that current tools rarely integrate LLMs into applications developed for individuals with ASD. This absence is particularly striking given the demonstrated potential of LLMs to enhance personalization, contextual adaptability, and communication support in neurodiverse populations. In parallel, the current body of literature reveals a lack of web-based systems that concurrently integrate (i) multi-user collaborative functionalities, (ii) AI-assisted generation of

¹<https://fespau.es/>

²<https://autrade.info/>

³<https://www.autismocastillalamancha.org/>

personalized routines, and (iii) export capabilities in open, standardized formats to facilitate interoperability across platforms [12], [16], [17], [18]. The aim is to address this combined technological space while strictly adhering to ASD-specific design guidelines, thereby ensuring both functional innovation and user-centered accessibility. Addressing this shortfall represents a critical step toward advancing inclusive, adaptive, and interoperable digital solutions for the ASD community.

The associations involved in the design of this application, previously mentioned (FESPAU, AUTRADE, and FACLM), have maintained an active collaborative role throughout the development process, systematically reviewing successive prototypes and platform versions. This iterative and incremental approach [19], based on continuous validation, enabled prioritizing key functionalities such as the inclusion of a shared calendar, automatic plan translation, and the implementation of an invitation system.

Thus, the objective of this work is to describe PlanTEA-WM, a web platform that, through the integration of multi-user functionalities and AI-powered tools, offers a robust, accessible, and collaborative solution for routine planning in individuals with ASD.

The rest of this article is structured as follows. Section II reviews related work and supporting tools. Section III then provides a detailed description of the main functionalities of PlanTEA-WM. Section IV describes the system architecture, detailing the technologies used and the application's structure. Section V reports the preliminary validation process and the accessibility-compliance checks carried out. Section VI illustrates, through a representative use scenario, the practical functionality of the application in a real-world context. Section VII discusses the results of the platform evaluation. Finally, Section VIII discusses the conclusions of this work, the ethical impact of AI integration, and possible future research directions.

II. RELATED WORK

In the field of technological tools designed to support individuals with ASD, various applications focus on routine planning and communication through pictograms [20]. Recent academic prototypes reinforce this trend, but they still cover the problem space only partially. A systematic review of 250 commercially available mobile applications targeting users with ASD [21] revealed that the vast majority are limited to AAC-based communication boards and elementary educational games; that only a small subset incorporates AI-driven functionalities; and that almost none offer support for data export or interoperability through open formats.

Below, some of the most relevant commercial solutions are described:

- **Proloquo2Go**⁴: An AAC application for iOS and macOS devices, designed for individuals with speech difficulties. It offers an extensive vocabulary based on

pictograms and predictive text, with multilingual support and customization options. Its interface allows users to construct sentences dynamically, enabling effective communication. However, it does not incorporate a routine planning system, calendar, or advanced multi-user collaboration functionalities.

- **ThruDay**⁵: A free visual planning application available on web, iOS, and Android, designed for neurodivergent individuals, including those with ASD, Attention Deficit and Hyperactivity Disorder (ADHD), dyslexia, or epilepsy. It supports daily routines via visual task lists, note-taking, and journaling, and incorporates a three-color (green–yellow–red) self-report scale that promotes emotional awareness. Assistants (e.g., family or professionals) can be invited to co-plan and receive real-time notifications. ThruDay presently lacks text-to-pictogram translation and does not export routines to open formats such as PDF or XML.
- **Tiimo**⁶: A cross-platform visual planner aimed at individuals with ASD or ADHD. Tiimo offers customizable schedules with icons and color coding, a focus timer, and an AI-assisted agent that decomposes tasks and proposes realistic time blocks. Cloud synchronization ensures data consistency across devices. Automatic pictogram generation and standards-based export are not yet provided.
- **Structured**⁷: An iOS, Android, and web application that unifies calendar events, to-do items, and habit tracking on a single chronological timeline. Subtasks, notes, and an “energy monitor” help balance cognitive load, while personalized colors and icons aid engagement. The platform omits text-to-pictogram conversion and PDF/XML export capabilities.
- **Morgen**⁸: A cross-platform productivity suite for Windows, macOS, Linux, iOS, and Android that unifies calendar management, task organization, and scheduling in a single interface. It integrates multiple calendar services—including Google Calendar,⁹ Outlook,¹⁰ iCloud,¹¹ and CalDAV¹²—so users can consolidate disparate calendars into a single view. Morgen embeds task-management features that let users create, organize, and schedule tasks directly on the calendar. Time-blocking facilitates the allocation of dedicated work slots, while personalized booking links and availability sharing streamline meeting coordination. Despite these strengths, Morgen does not support text-to-pictogram

⁵<https://thruDay.com/>

⁶<https://www.tiimoapp.com/>

⁷<https://structured.app/>

⁸<https://morgen.so/>

⁹<https://calendar.google.com/>

¹⁰<https://www.microsoft.com/en/microsoft-365/outlook>

¹¹<https://www.icloud.com/>

¹²CalDAV is an open standard protocol that allows clients to access scheduling information on a remote server. It is based on WebDAV and iCalendar, enabling calendar synchronization and event management across platforms. For further details, see <https://datatracker.ietf.org/doc/html/rfc4791>

⁴<https://www.assistiveware.com/es/productos/proloquo2go>

translation or export routines to open formats such as PDF or XML.

- **Habitica**¹³: An open-source habit-building and productivity app that gamifies task completion with role-playing game mechanics (avatars, experience points, quests). The reinforcement scheme can motivate some ASD users, yet Habitica lacks visual schedules, pictograms, and standard plan-export options.
- **BoardMaker**¹⁴: A commercial toolset for creating AAC materials such as communication boards and visual schedules. Boardmaker 7 combines a desktop editor with Boardmaker Online, giving access to a cloud library of templates, pictograms, and more than 120 000 community-shared activities. Educators can import their own photos, collaborate by sharing activities, and export projects—including visual schedules—as PDFs. The license cost and the need for a desktop editor still limit accessibility compared with fully web-based platforms.

Table 1 presents a comparison of the main functionalities supported by the analyzed applications. It can be observed that Proloquo2Go and Boardmaker remain the leading AAC editors, each providing very large symbol vocabularies. Thruday, Tiimo, Habitica, and Boardmaker all offer basic multi-user collaboration—through real-time assistant invitations (Thruday), shared profiles (Tiimo), guild/party task sharing (Habitica), and cloud activity sharing in Boardmaker Online. Four products now embed AI-assisted scheduling or task generation (Thruday, Tiimo, Structured, and Morgen), while Structured excels at blending calendar events, tasks, and habits on a single timeline, and Morgen specializes in aggregating heterogeneous calendars with time-blocking and booking links. However, none of the surveyed tools yet combine automatic text-to-pictogram translation, AI-based routine creation, standards-based export (PDF/XML), and a shared cloud workspace in a single package.

Beyond these tools, only a handful of research applications have begun to experiment with artificial-intelligence-driven personalization or dynamic routine generation. A recent study proposes a deep-learning recommendation engine that detects distraction or anxiety in real time and adapts the next pictogram step accordingly, illustrating the promise of AI yet stopping short of offering multi-user workflow, format export, or web deployment [16].

The new proposal, PlanTEA-WM, integrates in a single system the creation of schedules manually and automatically, by means of generative AI tools. For this purpose, it offers, among other functionalities, translation from text to pictograms, and includes external multimedia resources (pictograms, real images, etc.). Once the planning has been created, it can be exported to various formats (XML, PDF). In addition, the user with ASD can visualize it whenever he/she wants as a training to face situations that provoke

anxiety, as well as a support during the exposure to the anxiogenic situation. All of this is provided within the context of a multi-user environment, connecting *planners* with individuals with ASD.

In summary, commercial AAC and planning tools provide valuable capabilities such as rich symbol vocabularies and cross-device calendars, but they typically lack automatic text-to-pictogram translation and standards-based plan export within a multi-user, web-based workspace. Recent academic efforts have begun to explore AI-driven personalization and routine generation [12], [16], [18]. However, they often fall short of integrating multi-role collaboration, interoperable exports, and deployment as a single end-to-end web system. PlanTEA-WM is designed to address this specific combination.

III. PLANTEA-WM FUNCTIONALITIES

PlanTEA-WM offers a set of key functionalities that have been designed and validated based on the expertise of professionals from the collaborating associations, as well as the evaluation results of its predecessor, PlanTEA [15]. Consequently, PlanTEA-WM retains the features and functionalities of PlanTEA, whose effectiveness and usefulness have been verified, ensuring a smooth transition between applications:

- **Manages two user roles:** the *planner*, who creates and manages plans, and the *planned user* (person with ASD), who follows and visualizes the activities specified in the plans.
- **Supports event planning and management.** A sequence of pictograms representing a situation or activity is referred to as a *plan*. A plan scheduled for a specific date and time is called an *event*.

However, PlanTEA-WM introduces significant differences that expand the capabilities of the original application, such as multi-user management, AI-assisted planning tools, and enhanced accessibility across multiple devices. The following subsections describe its most notable functionalities.

A. USER MANAGEMENT

PlanTEA-WM introduces a multi-user model, allowing each registered *planner* to create and manage as many *planned users* (individuals with ASD) as needed and associate them with their account, as illustrated in Fig. 1. Additionally, through an invitation system, *planners* can share information about their *planned users* with other *planners*. To do so, they simply need to provide the recipient *planner*'s email address, and the application generates a unique link sent via email using the Brevo¹⁵ API. The recipient can either accept or decline the invitation. If accepted, both *planners* gain shared management of the same *planned user*.

Invitation links are implemented as single-use tokens generated by the backend and associated with the recipient's email address. When the link is accessed, the backend

¹³<https://habitica.com/static/home>

¹⁴<https://tecnioaccessible.net/catalogo/boardmaker>

¹⁵<https://www.brevo.com/es/>

TABLE 1. Comparison of functionalities supported by the analyzed applications.

Main Supported Functionalities	Application						
	Proloquo2Go	ThruDay	Tiimo	Structured	Morgen	Habituica	BoardMaker
Medical context	✓	X	X	X	X	X	✓
Plan creation	X	✓	✓	✓	✓	✓	✓
Plan presentation	X	✓	✓	✓	✓	✓	✓
Plan management	X	✓	✓	✓	✓	✓	✓
Add custom images	✓	X	✓	X	X	X	✓
Text-to-pictogram translation	X	X	X	X	X	X	✓
Pictogram search	✓	X	✓	✓	X	X	✓
Generative AI	X	✓	✓	✓	✓	X	X
Export as XML	X	X	X	X	X	X	X
Export as PDF	X	X	X	X	X	X	✓
Multi-user support	X	✓	✓	X	X	✓	✓

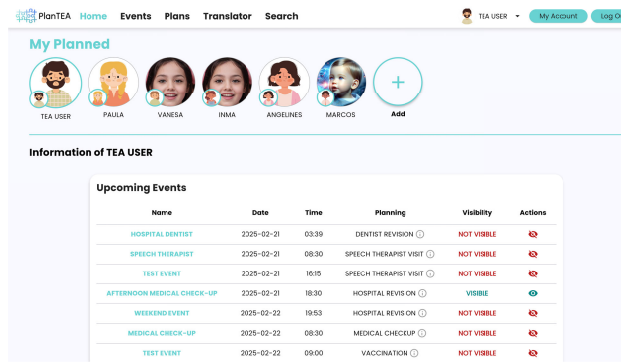


FIGURE 1. View of the dashboard available for the *planner* role, where the user can see, at the top, the list of their *planned users* (individuals with ASD) and the upcoming events associated with the selected user with ASD.

verifies that the authenticated account corresponds to the intended recipient; otherwise, sign-in is required before validation. Moreover, tokens are immediately invalidated upon acceptance or rejection. Once an invitation is accepted, the recipient receives the same permissions as existing members, including the ability to view, schedule, and manage events for the shared *planned user*.

Plans are always owned by their creator, who retains sole authority to modify their original content from the planning view. When an ASD user account is shared, all authorized *planners* can view every plan associated with that user and freely create or edit their own. However, if a *planner* edits a plan that does not belong to them from the calendar view, the system automatically generates a copy linked to them, ensuring the original remains unchanged. This access-control model preserves authorship integrity, supports traceability, and enforces a clear separation between view and edit permissions, thereby fostering secure collaborative workflows and minimizing the risk of data corruption or unauthorized changes.

This approach overcomes the limitations of the PlanTEA app, as information is now stored in a centralized and remote database (PostgreSQL), accessible from any Internet-connected device.

In the dashboard shown in Fig. 1, hovering over the information icon next to each plan displays a thumbnail

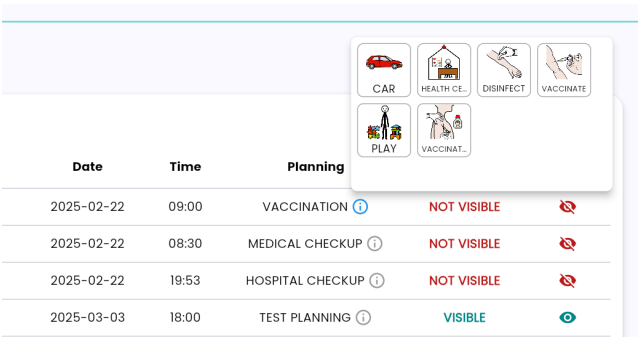


FIGURE 2. Thumbnail preview of pictogram sequences when hovering over the information icon in the dashboard.

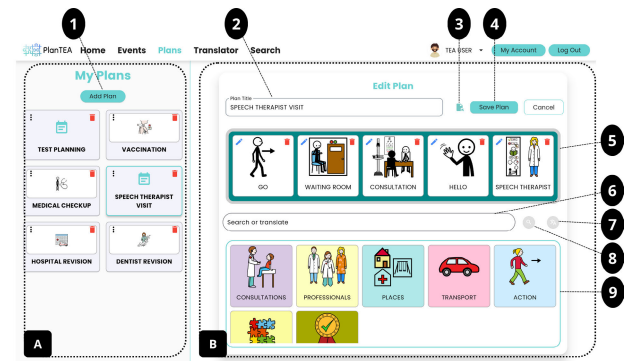


FIGURE 3. Plan management screen, which includes a list of plans on the left (A) with a button to create a new one (1), and a detailed view on the right (B), which includes: a title field (2); a button to import a previously created plan (3) or save it (4); an interactive board to add, modify, or delete pictograms (5); a field to enter a word or text (6) with options to search (8) or translate (7); and a collection of predefined pictogram categories (9).

preview of the pictograms that compose the plan. Fig. 2 illustrates this feature, allowing users to quickly visualize the pictogram sequence without accessing the detailed planning view. This functionality streamlines the management of events assigned to each *planned user*.

B. PLAN AND EVENT MANAGEMENT

The creation and editing of visual plans and routines are performed in a simple and flexible manner through the “Plans” view, as shown in Fig.3.

Such view follows the Master-Detail pattern, a design strategy that organizes information into two sections: a *master view* containing a list of items, and a *detail view* displaying specific information about the selected item. It is commonly used in applications handling large volumes of data to enhance navigation and user experience (UX) [22]. Accordingly, the left section (Fig. 3 (A)) functions as the master view, showing a list of existing plans along with a button (Fig. 3.1) to create new ones. The right section (Fig. 3 (B)), serving as the detail view, is dedicated to editing the selected plan or creating a new one. In addition to assigning an identifying title (Fig. 3.2), the *planner* can perform various actions to define routines: adding and reordering pictograms by dragging them into the desired position within the sequence (Fig. 3.5); saving changes to the plan in the system (Fig. 3.4); importing plans from XML (Fig. 3.3); exporting them to PDF format for later reuse; and searching (Fig. 3.8) or translating (Fig. 3.7) the pictograms associated with an entered word or text (Fig. 3.6).

Pictograms can also be chosen from the predefined categories provided (Fig. 3.9), all sourced from ARASAAC. The selection and organization of the categories were specifically designed to support the needs of individuals with ASD in medical and health-related contexts, which represent a particularly sensitive and recurrent scenario in the daily lives of these users. By focusing on health environments, the platform facilitates preparation and understanding before, during, and after medical appointments, thus reducing uncertainty and anxiety associated with such events.

The predefined categories include:

- **Consultations:** This medically oriented category groups pictograms according to common types of medical appointments. It includes subcategories such as *Revision*, *Vaccination*, *Tests*, *Ophthalmology*, *Dentistry*, and *Speech Therapy*. Each subcategory contains visual symbols representing actions, items, or procedures specific to that type of consultation, supporting comprehension and predictability for the user.
- **Professionals:** Contains pictograms representing key healthcare personnel, including *Doctor*, *Speech Therapist*, and *Nurse*, supporting recognition and familiarization with individuals involved in the user's care.
- **Places:** Focuses on locations where healthcare activities occur, such as *Health Center*, *Hospital*, and *Home*, providing spatial context within routines and events.
- **Transport:** Includes pictograms for common modes of transportation like *Car*, *Bus*, and *Ambulance*, which are often used to reach medical facilities or related environments.
- **Action:** Offers general-purpose verbs such as *Go*, *Return*, *Wait*, *Enter*, and *Greet*, which are essential for constructing simple, understandable sequences within visual plans.
- **Entertainment:** Though not strictly medical, this category includes *Play* and *Read*, supporting the inclusion

of leisure activities in daily schedules or to provide moments of relaxation surrounding healthcare events.

- **Reward:** Designed to represent positive reinforcement, this category includes pictograms such as *Play*, *Go to the Park*, and *Watch TV*, commonly used to motivate and support cooperative behavior during medical routines.

These categories enhance the platform's ability to create clear and contextually appropriate visual plans, improving user comprehension, reducing emotional stress, and fostering autonomy and communication skills.

Other possibilities available to the *planner*, which were particularly well received by experts from the collaborating associations, include renaming pictograms already inserted in the sequence (Fig. 3.5) or replacing them with real images of familiar places, people, or objects [23].

Additionally, there is an "Events" view, as illustrated in Fig. 4, which contains a monthly calendar (Fig. 4.1) highlighting the days with associated events, as well as a list of scheduled events (Fig. 4.2) with editing options. Through this list, it is possible to define the single "visible" event for the *planned user* (Fig. 4.3) so that its interface only displays the plan associated with the event selected by the *planner*. Events can be created directly from the calendar (Fig. 4.1) by selecting a specific day or by pressing on the specific button (Fig. 4.4), which opens the view for adding a new event to the calendar (Fig. 4 (B)).

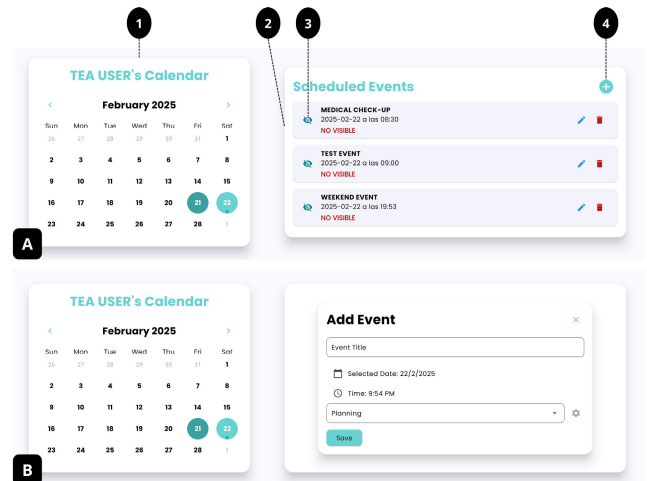


FIGURE 4. (A) Calendar and events view, allowing users to check the list of activities for the selected day. (B) View for creating a new event.

C. PICTOGRAM SEARCH AND TRANSLATOR

PlanTEA-WM incorporates two highly powerful functionalities for pictogram processing and assisted plan creation:

- **Pictogram search engine:** The *planner* can enter a keyword, and the application connects in real time to the ARASAAC API,¹⁶ retrieving all related pictograms.

¹⁶The ARASAAC API provides programmatic access to ARASAAC's pictographic resources, enabling developers to search, retrieve, and integrate pictograms into their applications. It facilitates the creation of accessible communication tools for individuals with speech and language difficulties. More information is available at: <https://beta.arasaac.org/developers/api>

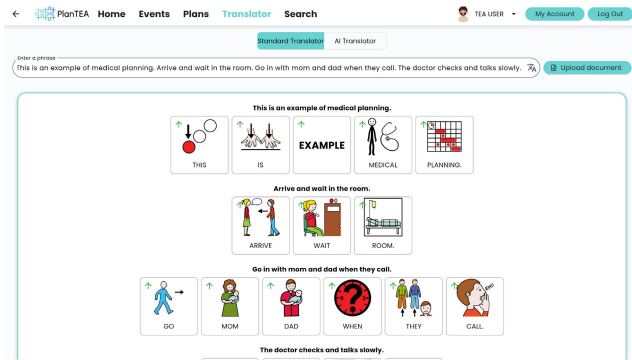


FIGURE 5. Example of translation from text to pictograms.

This functionality is particularly useful for quickly finding the image associated with a place, person, object, or action.

- **Text-to-pictogram translator:** The application allows users to introduce a full sentence in natural language as input (Fig. 5). The translator processes each word by removing irrelevant punctuation marks and applying lemmatization¹⁷ for words that do not have a direct match [24]. For example, if the user types “*Juan is playing in the park*”, the application processes the sentence and retrieves the associated pictograms, resulting in “*Juan is playing park*”. The implemented lemmatization mechanisms improve accuracy when the original word is conjugated or in plural form, preventing empty results. Additionally, the translator interface provides an ‘Upload document’ functionality that allows users to upload a text file and have its contents automatically converted into pictograms. This streamlines the process for longer or previously prepared texts, eliminating the need to manually copy and paste large amounts of information.

These tools eliminate the need to manually search for each pictogram, significantly speeding up the creation of routines and plans.

D. INTEGRATION OF GENERATIVE ARTIFICIAL INTELLIGENCE

One of the most innovative features of PlanTEA-WM is the incorporation of generative AI functionalities to automatically suggest routines [17], [18]. Its goal is not to replace the *planner*’s role but to complement it, saving time in the creation of plans and providing creative options that may be helpful.

The planning interface enables the *planner* to formulate a natural-language request, for instance, “*Simple routine for brushing teeth*”. This input is processed by a LLM hosted on OpenRouter,¹⁸ which generates an adapted textual output

¹⁷Lemmatization is the process of reducing a word to its base or *lemma*, considering its meaning and grammatical context. Unlike *stemming*, which merely removes suffixes, lemmatization applies linguistic rules to produce complete, meaningful words.

¹⁸<https://openrouter.ai/>

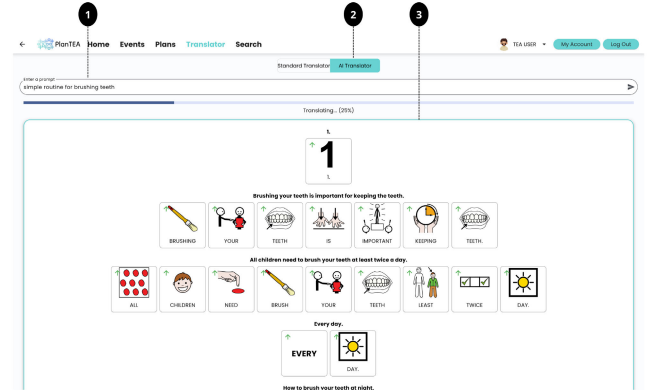


FIGURE 6. Generative AI translator (2) providing a pictogram sequence (3) corresponding to the entered text (“prompt”): “*simple routine for brushing teeth*” (1).

optimized for brevity, clarity, and the use of concrete, context-relevant vocabulary. The resulting text is then automatically transformed into a pictogram sequence via a dedicated text-to-pictogram translation module, producing a multimodal representation aligned with the communicative needs of individuals with ASD. The *planner* retains editorial control over the generated sequence, with the capacity to delete, add, or rearrange pictograms to better reflect situational constraints and the specific cognitive or linguistic profile of the target user (Fig. 6 illustrates an example derived from the aforementioned prompt).

To systematically ensure clarity, semantic precision, and accessibility, each planner-provided prompt is automatically prefixed with a system-level pre-prompt. This pre-prompt enforces constraints such as the generation of short, literal, stepwise instructions; the avoidance of figurative or idiomatic language; and the selection of vocabulary calibrated to the comprehension levels of ASD individuals. The LLM output is presented as an editable draft, subject to mandatory human review and approval prior to being saved or linked to an event. This human-in-the-loop validation mechanism is consistent with best practices in assistive technology design, ensuring both semantic fidelity and user-centered appropriateness. By integrating AI-driven content generation with structured human oversight, the system aligns with contemporary HCI principles, fostering reliability, transparency, and inclusivity in technology-mediated communication support.

E. PLANNED USER VIEW

While the *planner* has access to a full control panel, as described in Sections III-A to III-D, the *planned user* interface only displays the event marked as “visible” by their *planner* (Fig. 3). The view provided by PlanTEA-WM for event visualization is designed to be simple and allows ASD users to simulate the execution of the steps that make up the plan. This is achieved using play, forward, and backward options, as shown in Fig. 7.

The event can be simulated by pressing the play button. Each time a step is completed, its pictogram turns gray to

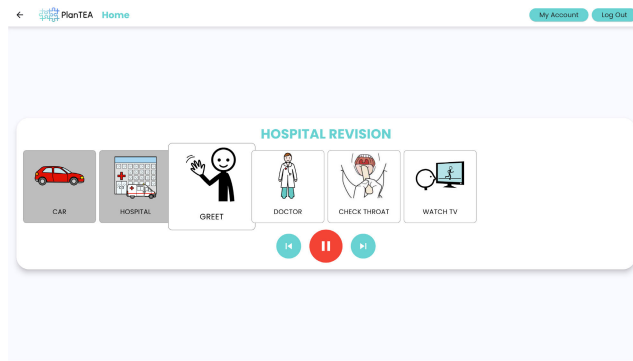


FIGURE 7. View available for the *planned user*, displaying the event plan in a step-by-step format. The first two pictograms are gray, indicating they have been completed. The current pictogram increases in size, while the remaining steps retain their original size and color.

distinguish it from those yet to be completed, providing visual feedback. Meanwhile, the pictogram representing the current step increases in size, while the remaining pictograms (pending steps) retain their original size and color.

IV. SYSTEM ARCHITECTURE

During the development of PlanTEA-WM, an iterative and incremental approach was followed, with frequent releases that allowed each new functionality to be validated alongside ASD professionals from the collaborating associations [25]. In each iteration, review meetings, the creation of low- and high-fidelity prototypes, and usability testing were conducted [26]. This process facilitated quick adjustments to requirements, ensuring that the solution continuously adapted to the real needs of end users.

The architecture of PlanTEA-WM adheres to a three-tier design pattern, comprising the presentation, application, and data layers, which correspond respectively to the *frontend* (the visible part of an application with which the user interacts, responsible for presentation and user experience), the *backend* application logic (managing data processing and communication with external services), and the persistence layer (responsible for storage and queries). This layered structure facilitates modularity, maintainability, and scalability, enabling a clear separation of concerns between user interface rendering, domain-specific business logic, and data management operations. The *frontend* is implemented using Flutter Web,¹⁹ following an MVVM (Model-View-ViewModel) pattern, which ensures a clear separation between presentation logic and state management. The *backend* is built on Node.js²⁰ and Express,²¹ adopting an MVC (Model-View-Controller) model, where business logic and persistence services are encapsulated. Data storage is handled by the relational database PostgreSQL,²² which facilitates scalability and multi-user collaboration.

Fig. 8 presents the component diagram of the application, illustrating how these components integrate with external APIs: ARASAAC, for accessing the pictogram database; OpenRouter, for incorporating generative AI functionalities; and Brevo, for the automatic sending of emails (such as invitations and shared plans). This results in a flexible architecture, designed to incorporate new features and support collaboration between multiple *planners* and users with ASD.

At the implementation level, the system realizes these layers as follows: (i) frontend views and view-models in Flutter Web, (ii) application services in Express (controllers and domain services for plans, events, invitations and AI requests), and (iii) data access components, including repositories and SQL queries executed over PostgreSQL. External integrations with OpenRouter and Brevo are encapsulated within backend service adapters, ensuring that the presentation layer remains decoupled from specific data sources and third-party APIs. In contrast, ARASAAC, as a public API that does not require authentication, is accessed directly from the frontend to optimize latency and reduce backend load. This design balances architectural decoupling with performance considerations.

The integration of the LLM module follows a clear workflow, as illustrated in Fig. 9. The *frontend* sends the user prompt to the *backend*, which appends a predefined system pre-prompt and forwards the request to the selected model through OpenRouter. Partial responses are streamed back to the client, where sentences are incrementally reconstructed. Once a full sentence is detected, it is normalized (removing stop words and punctuation, and applying lemmatization if needed) and each word is queried against the ARASAAC API. The resulting pictograms are progressively displayed in the correct order, ensuring that the *planner* can follow the translation step by step. The final pictogram sequence is stored only after explicit confirmation by the *planner*.

In addition, as part of the generative AI integration (Fig. 8), a preliminary comparative evaluation of three different language models was conducted: DeepSeek V3,²³ Gemma 3 1B IT,²⁴ and Meta LLaMA 3.3 70B Instruct.²⁵ Five prompts representing habitual scenarios for individuals with ASD were selected: (1) brushing teeth, (2) catching a bus, (3) planning a hospital visit followed by a reward at a park, (4) techniques to calm down when angry, and (5) asking for help at school. For each prompt, a dedicated rubric was used to assess: clarity of language, suitability for ASD (literal style and step-by-step structure), relevance to the requested context, and overall alignment with the prompt's purpose. Additionally, both the initial delay and total response time were measured using a Python script to evaluate practical performance. While this evaluation provided useful insights for selecting the current model, a more comprehensive study

¹⁹<https://flutter.dev/multi-platform/web>

²⁰<https://nodejs.org/es>

²¹<https://expressjs.com/>

²²<https://www.postgresql.org/>

²³<https://openrouter.ai/deepseek/deepseek-chat:free>

²⁴<https://openrouter.ai/google/gemma-3-1b-it:free>

²⁵<https://openrouter.ai/meta-llama/llama-3.3-70b-instruct:free>

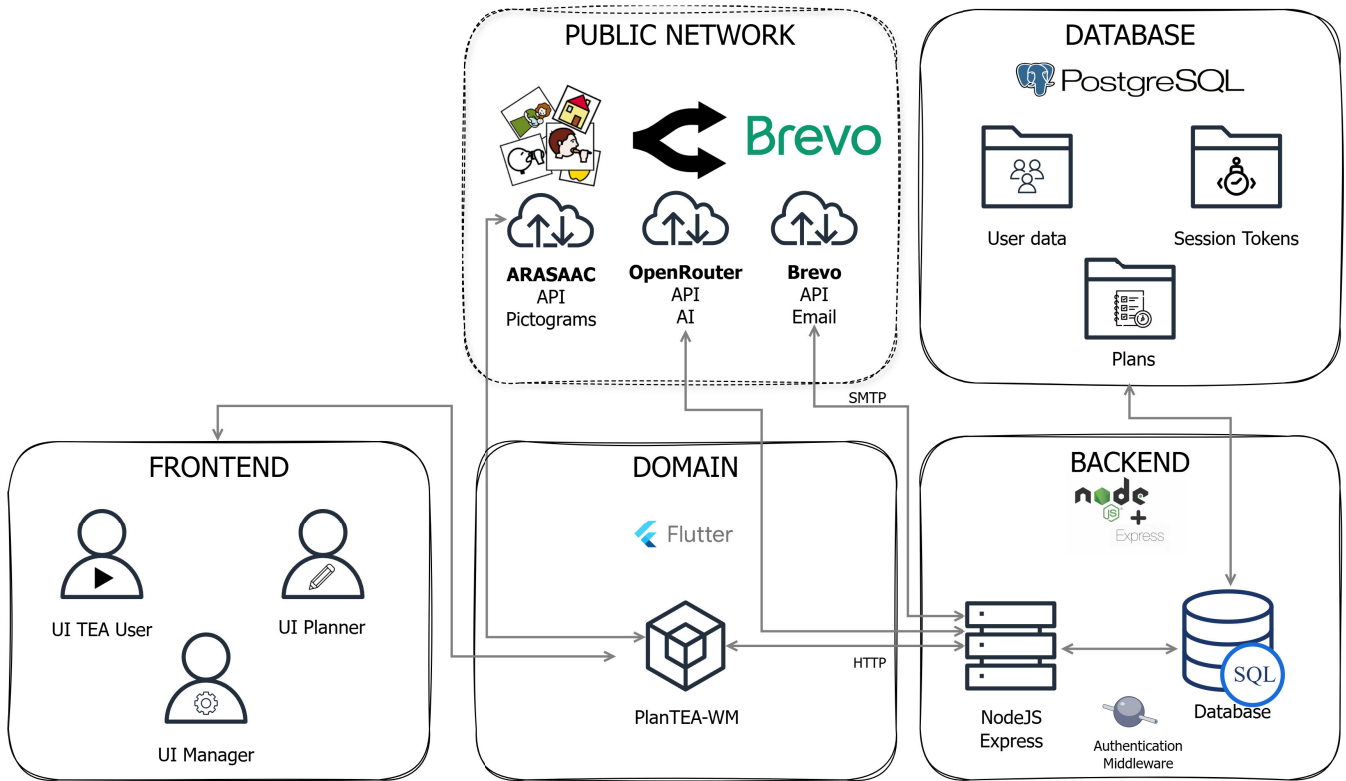


FIGURE 8. General overview of PlanTEA-WM architecture.

is being prepared to systematically access multiple LLMs in ASD and contexts for potential integration into PlanTEA-WM.

Based on these preliminary tests, DeepSeek V3 consistently delivered clear, well-structured instructions, without incurring significant generation delays. Its responses often achieved high scores on the rubric—particularly in clarity and literal adaptation—making it the most suitable option for PlanTEA-WM. This model strikes an effective balance between the simplicity required by users with ASD and the fluidity needed for real-time usage.

Consequently, the PlanTEA-WM architecture combines the flexibility of its different components with a robust generative model tailored to the communication needs of individuals on the autism spectrum.

V. FUNCTIONAL, USABILITY, AND ACCESSIBILITY VALIDATION

This section describes tests conducted to ensure that PlanTEA-WM is a robust, user-friendly, and accessible application for its target audience.

A. USABILITY AND FUNCTIONAL TESTING

Preliminary usability assessment focused on verifying menu clarity, ease of creating new plans, and simplicity in routine playback. Additionally, feedback from ASD professionals

was collected, validating that the interface was perceived as intuitive and easy to navigate.

Moreover, a heuristic inspection was conducted following the ASD-oriented property checklist proposed by Valencia et al. [27], which defines nine UX categories tailored to the needs and characteristics of individuals with ASD. This checklist provided a structured framework to verify that the platform's interface elements aligned with specific usability and user experience recommendations for this population.

General functionality tests were also carried out, ensuring the correct interaction between the *frontend* and *backend* (e.g., user registration, event creation). No critical errors were recorded, though adjustments were made to field validation and error messages.

B. ACCESSIBILITY HEURISTIC AND COLOR CONTRAST CHECK

Throughout the development of PlanTEA-WM, various accessibility design and evaluation strategies were applied, following the Web Content Accessibility Guidelines (WCAG 2.1) established by World Wide Web Consortium²⁶ (W3C). Additionally, design heuristics specifically aimed at users with ASD were considered [23], [28].

The main accessibility principles considered include perceptibility, operability, comprehensibility, and predictability,

²⁶<https://www.w3.org/>

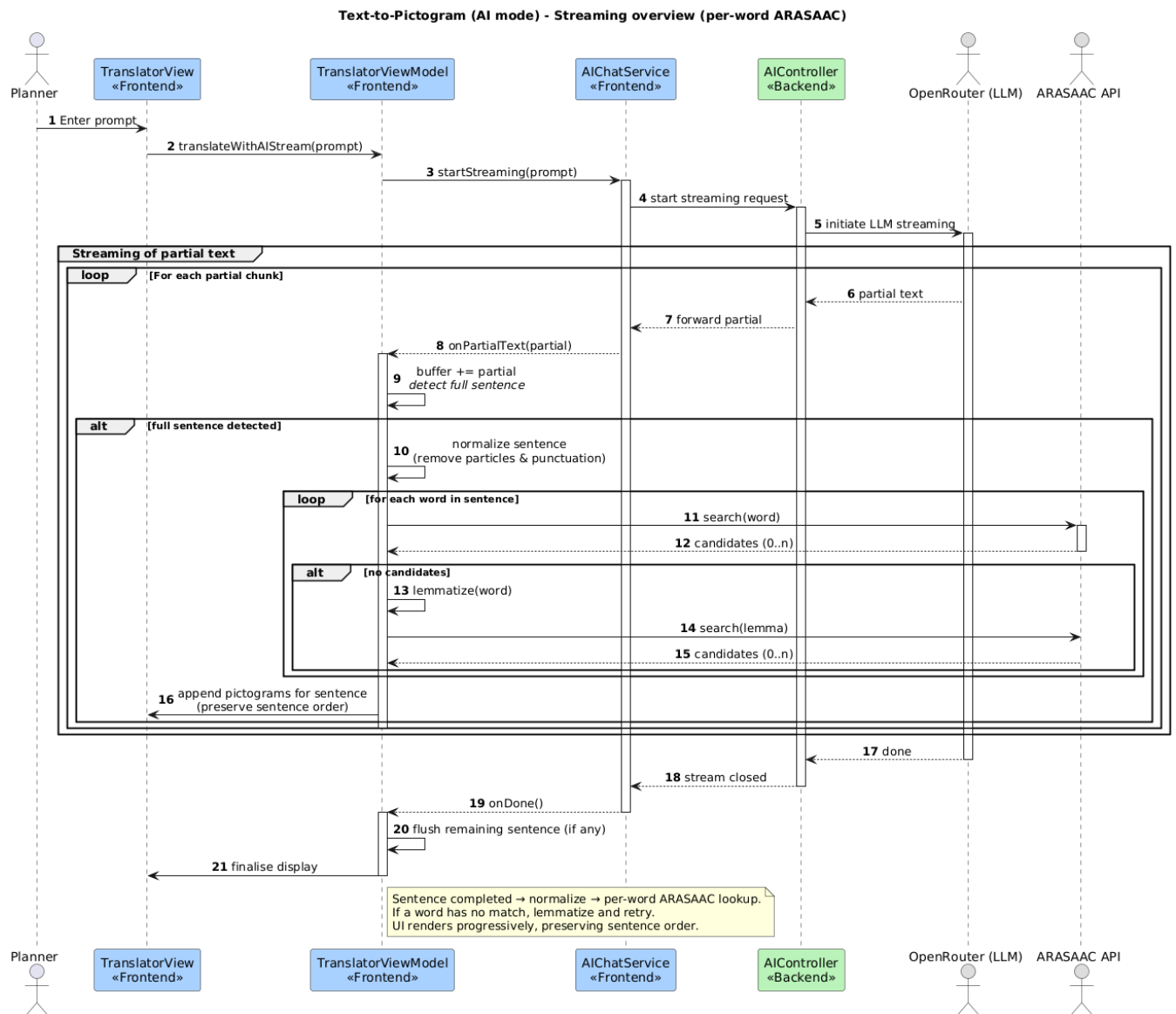


FIGURE 9. Sequence diagram of LLM-based text-to-pictogram translation in PlanTEA-WM.

ensuring that information is easy to find, read, understand, and interact with for all users [29]:

- **Perceptibility:** Proper contrast between text and background was prioritized, consistently meeting AA and even AAA levels in critical elements (e.g., titles and labels). Pictogram size and readability were also reviewed to ensure that individuals with visual impairments or color blindness can distinguish them properly.
- **Operability:** The interface design encourages simple navigation and interaction through basic actions (click, drag, tap), avoiding complex gestures. The application is structured to be fully keyboard-accessible, benefiting users with reduced mobility.
- **Comprehensibility and predictability:** To reduce cognitive load, consistent menus and clear language (free of

unnecessary jargon) were used. Additionally, a repetitive structure was maintained in screen and button layouts, enhancing consistency and anticipation of actions.

As a key part of this validation, an exhaustive color contrast check was performed across the main screens. Specifically, contrast ratios were assessed using WhoCanUse,²⁷ and different color palettes were tested under protanopia²⁸ and protanomaly²⁹ conditions. For example, some buttons

²⁷<https://www.whocanuse.com/>

²⁸Protanopia is a type of color blindness in which the perception of red is absent due to the lack of red-sensitive cone cells in the retina. This results in difficulties distinguishing between red and green tones.

²⁹Protonomaly is a form of color blindness where sensitivity to red is reduced, causing reddish tones to appear more muted or to be confused with other colors, particularly green and brown.

initially had insufficient contrast (1.79:1), which was adjusted to achieve a ratio close to 9:1.

Overall, the adopted measures have ensured that the application interface achieves high levels of clarity and robustness, benefiting both users with ASD and individuals with visual impairments. As a result, PlanTEA-WM not only meets formal accessibility standards but also provides a more inclusive user experience, tailored to the real needs of its intended users. Beyond color and layout verification, a checklist-based audit was performed, covering keyboard navigation, including tab order and focus visibility within dialogs, lists, and buttons. As future work, a formal expert accessibility evaluation against WCAG 2.1/2.2 will be carried out, including comprehensive screen-reader testing.

VI. USE SCENARIO: PREPARING FOR A PEDIATRIC CHECK-UP USING PLANTEA-WM

This section describes a hypothetical use scenario and how PlanTEA-WM can be used to support it. Suppose Raul is a 12-year-old child with level 2 ASD, exhibiting significant difficulties in social interaction, and often displaying repetitive or restrictive behaviors that considerably interfere with daily life.

Raul needs to visit the dentist in a few days for a routine check-up and teeth cleaning. His mother, Elisa, will use PlanTEA-WM for the first time to help him anticipate the situation. The following steps illustrate how this scenario can be supported using PlanTEA-WM.

- 1) **Account creation and role assignment.** Elisa registers as a *planner* and enters Raul's basic details to associate him as her *planned user*. These include the user's name, password, optional email address, avatar, and profile image. Two additional fields are also configured: a soothing activity, which refers to a visual, sensory, or tangible element used to reduce anxiety and provide reassurance to individuals with ASD in stressful situations [30]; and the pictogram display mode for the *planned user*, which can be set to text and image (default), image only, or text only, depending on the user's individual support needs.
- 2) **Plan design.** In the "Plans" section (Fig. 3), Elisa selects the "Add Plan" to start creating a step-by-step sequence for the dentist visit. Using the pictogram search engine, she selects images such as "wait", "examination", and "dentist", dragging them onto the sequence creation and editing board. Since Elisa has a photo of Marcos (Raul's dentist), she edits the pictogram to include his name and image (Fig. 10), helping Raul better anticipate the situation. The use of real and familiar images of people, places, or objects can significantly enhance comprehension and reduce uncertainty for individuals with ASD, as it aligns more closely with their everyday experiences and improves the predictability of upcoming events [31].

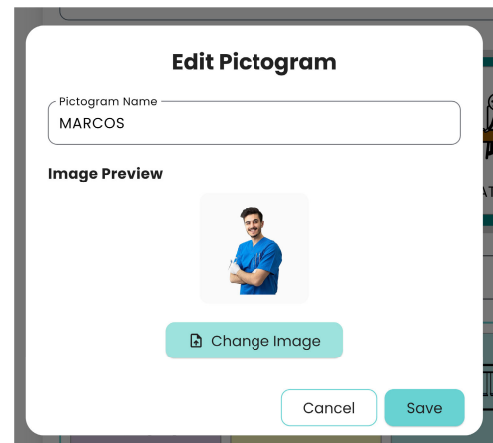


FIGURE 10. View allowing pictogram editing.

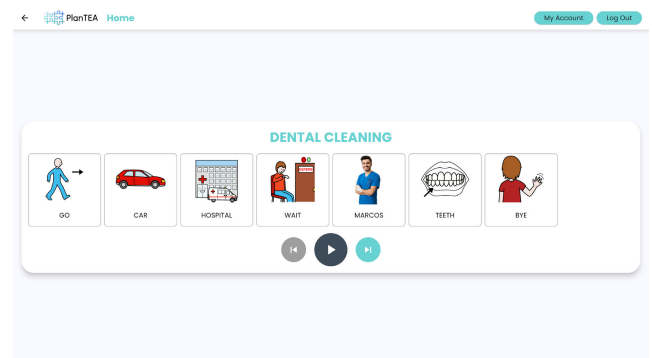


FIGURE 11. View available for Raul, showing the planning that Elisa has marked as visible.

- To explain to Raul that he will be having a dental cleaning, Elisa selects the "Translator" option from the menu and decides to use the AI-assisted translation function (Fig. 6.2), where she enters the text (Fig. 6.1) "How to get a dental cleaning". Once the final pictogram sequence is completed, Elisa saves it as a new plan under the name "DENTAL CLEANING".
- 3) **Assigning the plan to the calendar.** Elisa then selects the "Calendar" option (Fig. 4) and picks the date of Raul's dental check-up. Since there are no other planned events for that day, she opens the window to add a new event (Fig. 4 (B)), where she enters a name, selects the time, and chooses "Dental Cleaning" from the planning dropdown menu. Once saved, she marks it as "visible" for Raul in the event list (Fig. 4(A)).
- 4) **Routine anticipation by the user with ASD.** When Raul logs in (*planned user* role), he will only see the pictogram sequence associated with the "Dental Cleaning" plan (Fig. 11). With his mother's help, if needed, he can view the steps as many times as necessary before the visit to the dentist. Additionally, Raul can manually move forward and backward through the sequence using the integrated player, adjusting to his own pace and needs.

VII. PRELIMINARY EVALUATION OF PLANTEA-WM

A preliminary study was designed to gather data on the usefulness and acceptance of PlanTEA-WM. The goal was to collect information about its main features (multi-user support, control panel, plan previews, AI integration, etc.) and explore users' intention to adopt the tool. The following sections describe the process carried out.

A. STUDY OBJECTIVE AND RESEARCH QUESTIONS

To guide the evaluation and ensure a structured analysis of user feedback, the following three research questions (RQs) were defined:

- **RQ1.** Which PlanTEA-WM functionalities are considered the most useful?
- **RQ2.** What is the participants' perception of PlanTEA-WM's acceptance in terms of Perceived Ease of Use (PEU), Perceived Usefulness (PU), and Intention to Use (ITU)?
- **RQ3.** Which strong and weak aspects do participants report, and what improvement suggestions do they offer for PlanTEA-WM?

Answering these RQs helps identify the tool's strengths, limitations, and areas for enhancement. It also estimates how likely potential users (people with ASD, families, caregivers, healthcare professionals, educators, etc.) are to adopt it.

B. EXPERIMENT AND MEASUREMENT INSTRUMENT

The evaluation took place during an online seminar titled "Nuevas Tecnologías y TEA" ("New Technologies and ASD"), held in March 2025. This event was organized by an association of people with autism and their families (AUTRADE). It targeted individuals with ASD, their relatives, ASD experts, and healthcare professionals.

During the seminar, PlanTEA-WM was presented through a 15-minute demonstration of its main features and functionalities. Afterward, attendees were asked to complete an anonymous online questionnaire using Microsoft Forms.³⁰ A total of forty-four valid responses were obtained.

Regarding participant profile, most (33) were professionals who work directly with people with ASD (therapists or educators). Eleven were relatives, and 2 selected "Other" (volunteers or students collaborating with ASD organizations). No participant identified as a person with ASD. Some respondents had more than one role (for example, both professional and family member).

The questionnaire was structured in several blocks, covering the following dimensions:

- **Background and participant profile.** This included gender and connection with the ASD community.
- **Perceived usefulness of PlanTEA-WM features.** Seven items (FUNC1 to FUNC7) were included to assess aspects such as multi-user support, text-to-pictogram translation, and the integration of generative

AI, among others. These items were rated on a 5-point Likert scale (1 = not useful at all, 5 = very useful) [32]. This block addressed RQ1.

- **PlanTEA-WM acceptance.** The Technology Acceptance Model (TAM), proposed by Davis in 1989 [33], was applied. TAM is widely used to assess the acceptance of new technologies, suggesting that the Intention to Use (ITU) depends on Perceived Ease of Use (PEU) and Perceived Usefulness (PU). In this evaluation, ITU was measured through two items (ITU1 and ITU2) focused on personal intent to use the platform and intention to recommend it. Each statement was scored on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). This block addressed RQ2.
- **Open-ended questions.** Three questions gathered opinions on strengths, weaknesses, and suggestions for improving PlanTEA-WM. This block addressed RQ3.

This measurement instrument combined quantitative scales and open-ended questions. It collected information on the perceived usefulness and acceptance of PlanTEA-WM, as well as areas for enhancement. The next section presents the evaluation results and discusses them in relation to RQ1, RQ2, and RQ3.

C. DATA ANALYSIS AND RESULTS

This section summarizes the findings from the 44 collected responses. The results focus on two main areas: the evaluation of PlanTEA-WM's functionalities and the tool's acceptance based on the TAM framework. Lastly, it includes participants' comments on the tool's strengths and weaknesses, obtained from the open-ended questions.

1) RQ1. WHICH PLANTEA-WM FUNCTIONALITIES ARE CONSIDERED MOST USEFUL?

To address RQ1, the perceived usefulness of the seven functionalities shown in Table 2 (FUNC1 to FUNC7) was analyzed. The table shows descriptive statistics (mean, standard deviation, median, and mode) for each item.

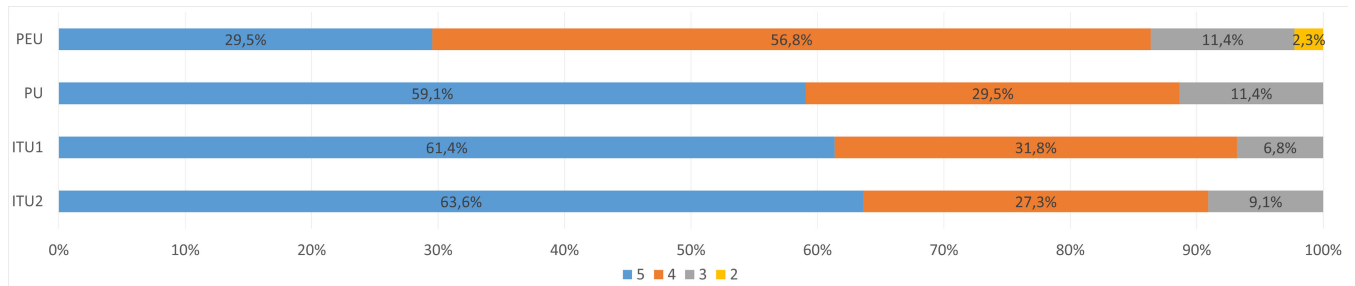
The results (Table 2) indicate that all functionalities were considered highly useful. Each item's mean exceeded 4.5 (on a scale of 1 to 5), with medians and modes of 5, and standard deviations ranging from 0.53 to 0.66. This suggests strong agreement among respondents. Multi-user collaboration (FUNC4) achieved the highest mean (4.75). Using PlanTEA-WM on multiple devices (FUNC1) also had a high mean (4.73), confirming that browser-based access is well-received.

Text-to-pictogram translation (FUNC5) and AI-generated planning suggestions (FUNC7) both had a mean of 4.68. These results underscore the importance of assistance and automation when creating plans. Although the thumbnail preview (FUNC3) had the lowest mean (4.55), it was still rated positively, indicating that the core features of PlanTEA-WM were well-received.

³⁰<https://www.microsoft.com/en/microsoft-365/online-surveys-polls-quizzes>

TABLE 2. Descriptive statistics of the main functionalities and characteristics of PlanTEA-WM.

Variable	Main functionalities and features of PlanTEA-WM	Mean	Standard Deviation	Median	Mode
FUNC1	Use of PlanTEA-WM through any web browser	4.73	0.54	5	5
FUNC2	Management panel for <i>planned users</i> and events	4.64	0.57	5	5
FUNC3	Thumbnail preview of plans in the management panel	4.55	0.66	5	5
FUNC4	Collaboration with others to plan for a person with ASD	4.75	0.53	5	5
FUNC5	Translation of text documents into pictograms	4.68	0.59	5	5
FUNC6	Insertion of pictograms into a plan from the translator	4.68	0.55	5	5
FUNC7	Use of generative AI to suggest plans	4.68	0.59	5	5

**FIGURE 12.** Stacked bar chart showing the percentage distribution of Likert scale responses for TAM dimensions (PEU, PU, ITU).**TABLE 3.** Descriptive statistics of the TAM dimensions (PEU, PU, and ITU) for PlanTEA-WM.

Variable	Item statement	Mean	Standard Deviation	Median	Mode
PEU	I think PlanTEA-WM is easy to use	4.14	0.69	4	4
PU	I think PlanTEA-WM is useful for planning and anticipating daily activities for individuals with ASD	4.48	0.69	5	5
ITU1	If it were available to me, I would use this application to plan and anticipate tasks	4.55	0.62	5	5
ITU2	I would recommend that other professionals use PlanTEA-WM	4.55	0.66	5	5

2) RQ2. WHAT IS THE PARTICIPANTS' PERCEPTION OF PlanTEA-WM's ACCEPTANCE IN TERMS OF PERCEIVED EASE OF USE (PEU), PERCEIVED USEFULNESS (PU), AND INTENTION TO USE (ITU)?

The second research question explores acceptance of the platform, measured through the TAM dimensions of PEU, PU, and ITU. Fig. 12 shows the distribution of responses by percentage for each Likert scale level. Table 3 summarizes the descriptive results, including the mean, standard deviation, median, and mode for each dimension.

The perceived ease of use (PEU) had a mean of 4.14. About 86% of respondents rated it 4 or 5, indicating a largely positive perception, though there was moderate variability (standard deviation 0.69). The perceived usefulness (PU) had a mean of 4.48, with 59.1% of participants giving the highest possible score. This suggests that they see PlanTEA-WM as a tool that can improve planning and task anticipation for ASD users.

Both ITU1 (personal intention to use) and ITU2 (intention to recommend) had a mean of 4.55 and a median of 5, with over 60% of responses at the highest rating. This highlights

strong willingness both to adopt PlanTEA-WM themselves and to recommend it to others. Overall, the TAM dimensions show good acceptance for PlanTEA-WM, with high ratings for usefulness and intention to use.

3) RQ3. WHICH STRONG AND WEAK ASPECTS DO PARTICIPANTS HIGHLIGHT, AND WHAT IMPROVEMENT IDEAS DO THEY SUGGEST FOR PlanTEA-WM?

Below is a qualitative analysis of the open-ended responses, which addresses research question RQ3. It covers the strong points, weak points, and improvement suggestions for PlanTEA-WM mentioned by participants.

Regarding the **strong points**, many comments praise the platform's ease of use and accessibility, using phrases such as "*its easy handling*" and "*very useful and simple*". The integration of AI stands out as a key differentiator. Participants mention automatic text-to-pictogram translation or quick plan generation ("*The AI part fascinated me, immediate creation*", "*Using artificial intelligence speeds things up a lot*"). They also appreciate the option to collaborate with different stakeholders (family members,

professionals), the event calendar, and how quickly one can create and edit plans. Several participants note “*many everyday uses*” and “*many possibilities*” for customizing activities to each individual’s needs, highlighting the system’s flexibility and its ability to “*improve communication with people with ASD*”.

Despite the overall positive reception, and even though some respondents said they found no weaknesses, a few **shortcomings** affect certain users. One issue is that PlanTEA-WM may seem complex to users with lower digital skills, such as older relatives or those with little technology experience [34]. Another concern is the need for “*voice support*” to reinforce how actions play out [35], along with fixing small AI errors (for example, translating “*pasta*” as food instead of toothpaste). Some participants note that the *planned user* role does not include any AAC functionalities, so they suggest adding a communication board to boost communicative autonomy [36]. Additionally, the large number of pictograms might confuse individuals with more severe ASD, making further personalization necessary for them. One proposed solution is to let users set preferences about the length of the AI-generated answer (“short”, “medium”, “long”). In this way, the *planner* can limit the number of pictograms shown, better tailoring the plan to each person’s needs.

Regarding **final suggestions**, participants propose adding two-way communication features for the ASD user [37]. Others stressed the importance of accommodating individuals with stronger cognitive or sensory challenges so that the tool is even more inclusive. Another idea is to include emoticon-like reactions for each event (for instance, a sad or happy face) to track how the user feels about the planned situation [38], [39]. In general, there is strong interest in seeing PlanTEA-WM continue to evolve and reach more settings, with remarks like “*Great job!*”, “*Wonderful proposal!*” and “*I would love to use it personally and with my colleagues at the center*”.

VIII. CONCLUSION AND FUTURE WORK

PlanTEA-WM on the original PlanTEA app to overcome its limitations, including single-user design and reliance on local storage, offering a comprehensive tool for routine planning and multi-user collaboration. Key features include role-based user management, calendar-based event management, pictogram search and translation, AI-assisted routine generation, and plan export/import in multiple formats. The platform is designed to be versatile, scalable, and suitable for family, clinical, and educational settings, with proven usability and accessibility aligned with WCAG guidelines. Preliminary accessibility validation has already been carried out, including systematic color-contrast checks, keyboard navigation, and checklist-based evaluations. As future work, an expert-based heuristic review and a formal WCAG 2.1/2.2 audit are planned, covering screen-reader compatibility and error-prevention mechanisms.

Generative AI accelerates routine creation by converting text sequences into pictograms, though planners must supervise outputs to ensure simplicity, clarity, and appropriateness while safeguarding privacy [40]. This supervision follows a human-in-the-loop approach, supported by constrained prompt templates and lightweight automatic checks to enforce step format and vocabulary adequacy. Future research will extend these mechanisms by exploring advanced prompt engineering techniques to improve adaptability and support individualized plan generation. The goal is to ensure that generated outputs are not only accurate but also aligned with the unique communication needs, comprehension levels, preferences and contextual factors of users with ASD.

Future work will also prioritize the design of adaptive personalization features to meet the varied needs of users, enabling fine-grained control over pictogram quantity, display configurations, and the integration of AAC boards or voice output. Such improvements are expected to increase the system’s scalability and broaden its applicability to diverse populations with communication support needs. Additional enhancements will include integration with external calendars, exploring synchronization with platforms like Google Calendar or Outlook, so that events created in PlanTEA-WM are automatically recorded and updated in each planner’s personal calendar.

The preliminary evaluation indicates high user satisfaction, particularly regarding AI support, rapid plan creation, and multi-user collaboration, while highlighting areas for improvement such as simplifying the learning curve and refining AI translations. Nevertheless, it is important to acknowledge the preliminary nature of the study. While feedback was highly positive regarding usability and acceptance, the evaluation has clear limitations: the relatively small sample size, the absence of individuals with ASD as direct participants, and the focus on perceived usefulness rather than actual effectiveness or long-term adoption. As future work, a more comprehensive evaluation will be conducted to address these gaps by increasing the participant pool, incorporating families and individuals with ASD, and complementing subjective perceptions with objective measures of effectiveness and sustained use. Such a longitudinal assessment will be essential to rigorously determine the platform’s real-world impact.

With these ongoing and planned enhancements, PlanTEA-WM aims to promote autonomy for individuals with ASD, support their families and caregivers, and extend its applicability to other populations requiring communication support, including those with intellectual disabilities, aphasia, Down syndrome, cerebral palsy, or age-related cognitive decline.

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of the evaluation sessions organized by AUTRADE, whose forthcoming assessment and insights will be crucial for validating and enhancing the usability and effectiveness of the platform in real-world scenarios.

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